

Guided Training of NeRFs for Medical Volume Rendering

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(a) Interactive view synthesis from a trained neural radiance field



(b) Subset of the training data

Figure 1: (a) Novel view synthesis during interactive exploration of an ex-vivo 60GB volume dataset using a compact neural radiance field. The model is trained on an optimized set of 60 pre-rendered views generated with Monte Carlo-based volumetric path tracing and (b) shows a subset of the training data. The Hierarchical Phase-Contrast Tomography (HiP-CT) brain scan is courtesy of the European Synchrotron Radiation Facility (ESRF) [Tafforeau et al. 2021].

ABSTRACT

Neural Radiance Fields (NeRF) trained on pre-rendered photorealistic images represent complex medical data in a fraction of the size, while interactive applications synthesize novel views directly from the neural networks. We demonstrate a practical implementation of NeRFs for high resolution CT volume data, using differentiable rendering for training view selection.

KEYWORDS

volume path tracing, neural radiance fields, differentiable rendering

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1 INTRODUCTION AND MOTIVATION

The described system addresses the challenge of visualizing very high-resolution medical images with modern photorealistic rendering techniques at interactive speeds. Computed Tomography (CT) scans on modern photon-counting systems are an example of a high-resolution imaging data source available for in-vivo clinical workflows, while ex-vivo imaging can produce 3D volume data hundreds of gigabytes in size. The proposed approach leverages the latest advances in neural radiance fields (NeRF) to train a compact neural representation from high quality volume rendering images and enable novel view synthesis in a range of applications, from education to procedure guidance. An optimization phase based on differentiable rendering creates a sparse set of views based on the contents of the medical scan, which allows for faster training of the neural model as part of interactive clinical applications.

2 TECHNICAL APPROACH

At the high level, NeRF view synthesis relies on the properties of a 3D scene encoded by a coordinate-based neural network rather

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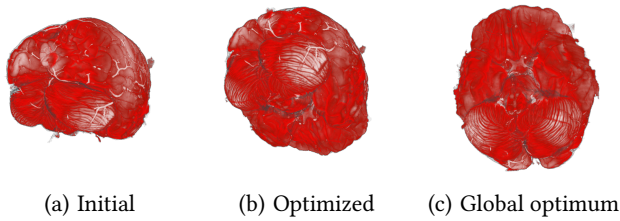


Figure 2: Gradient descent optimization using differentiable direct volume rendering and the visual entropy metric evolves the initial view (a) to show an increased amount of visual information (b). (c) is the overall global maximum.



Figure 3: View synthesis using a NeRF trained for 2,000 steps with optimized views (b) shows improved reconstruction of finer details compared to using the original training set (a).

than explicit rendering of 3D primitives [Xie et al. 2022]. Images are then produced by raycasting and inference on the trained neural model and various optimization techniques improve the performance to interactive levels suitable for many practical applications. Specifically, Müller et al. [Müller et al. 2022] describe an efficient implementation of both NeRF training and inference, which becomes the basis for the described system.

Differentiable rendering models the explicit relationship between the rendering parameters and the resulting images in traditional image synthesis. It obtains image-space derivatives with respect to the rendering parameters, which can be used in a variety of gradient-based optimization methods to solve inverse rendering problems. Our approach starts with segmentation of clinically important structures in the medical data and gradient descent optimization evolves a set of initial views based on differentiable rendering and the image-space visual entropy metric, which measures the amount of visual information in the current view. Mesh-based differentiable rendering [Loper and Black 2014] is used with explicit surface segmentation, while voxel masks and voxel classifications based on a transfer function are handled with differentiable direct volume rendering [Weiss and Westermann 2022].

3 RESULTS AND FUTURE WORK

The system targets the photorealistic visualization of very large 3D volume datasets, including photon-counting CT and Hierarchical Phase-Contrast Tomography (HiP-CT) images. The direct volume rendering employs a Monte Carlo path tracing algorithm with a

hybrid of volumetric scattering and surface-like scattering, modeled by phase functions and BRDFs respectively, and based on properties derived from the anatomical data [Kroes et al. 2012].

The example data is an ex-vivo 60GB brain volume dataset obtained from HiP-CT and resampled to a 50um resolution [Tafforeau et al. 2021]. The NeRF training images are generated at 2048x2048 resolution and 500 spp on NVIDIA A100 GPUs. A set of 60 cameras are positioned using stratified sampling of the volume bounding sphere and then a small number of gradient descent steps evolve each camera position to optimize the visual entropy of the image. The differentiable volume rendering uses a transfer function for the cerebral cortex and Fig. 2 shows the initial and final positions for a single camera. Fig. 3 compares the NeRF view synthesis quality, where using the training set with optimized views (Fig. 3(b)) reduces the pixel RMSE by 3% and the RMS of the perceptual hash difference by 58% compared to the path tracing results at 2,000 training steps. The reduction is 5% and 37% respectively at 20,000 training steps. Finally, Fig. 1(a) shows frames from the interactive training and view synthesis running on an NVIDIA RTX A5000 GPU at 33ms per 1080p frame, compared to 26 sec. for the 500 spp volume path tracing of the same image.

In a future extension, this type of view synthesis may be applied to augmented reality overlays for surgical guidance, where the view synthesis may combine NeRFs trained on images rendered from pre- and intra-operative scans of the patient, including segmentation and procedure planning graphical objects embedded in the volume data. Volume path tracing, NeRF training and view synthesis running in parallel is also of interest for interactive applications where the rendering parameters change as part of the clinical workflow and user inputs.

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The concepts and information presented in this paper are based on research results that are not commercially available. Future commercial availability cannot be guaranteed.

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